

BAULKHAM HILLS HIGH SCHOOL



MATHEMATICS ADVANCED ASSESSMENT

December 2012

Time allowed: 50 minutes + 5 minute reading time

STUDENT'S NAME: _____

TEACHER'S NAME: _____

General Instructions

- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question in section II
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

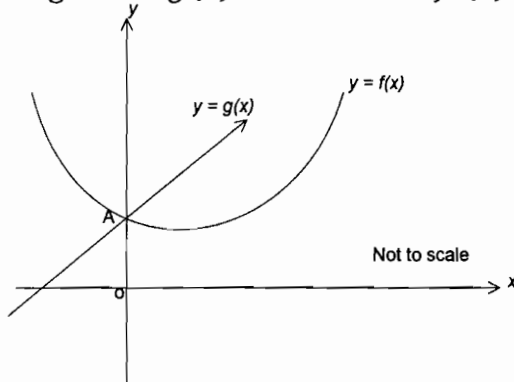
Topics Tested: *Integration, Series, Locus and Maximum, Minimum problems*

Section I - 4 marks

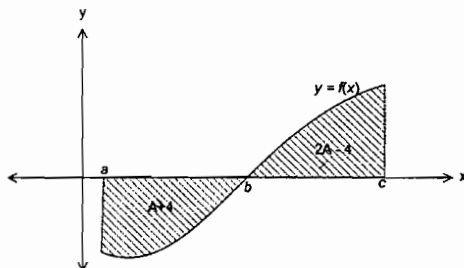
Use the multiple choice answer sheet for question 1-4

1. The n^{th} term of the sequence -1, 4, -9, 16, is
(A) 32 (B) $-2n$ (C) n^2 (D) $(-1)^n n^2$
2. Focus (1, -3), directrix is $x=5$. From the information given, the equation of the parabola is
(A) $(y+3)^2 = 8(x-3)$ (B) $(y-3)^2 = -8(x-3)$ (C) $(y+3)^2 = -8(x-3)$
(D) $(y-3)^2 = 8(x-3)$

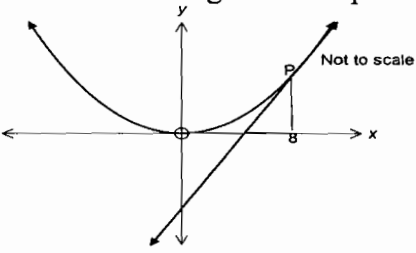
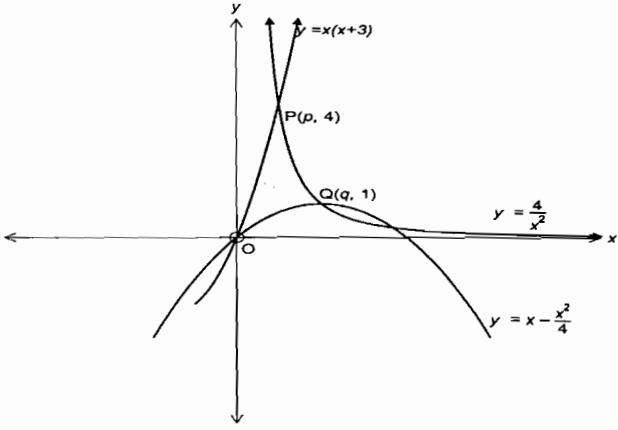
3. The graph of $y = g(x)$ and $y = f(x)$ intersect at the point A on the y -axis, as shown in the diagram. If $g(x) = 3x + 4$ and $f'(x) = 2x - 3$, find $f(x)$.



- (A) $x^2 - 3x + 7$ (B) $3x^2 + 4x + 4$ (C) $x^2 - 3x + 4$ (D) $x^2 + 4x - 7$
4. In the diagram shown below, the area between the curve $y = f(x)$, the x -axis, and the line $x = a$ is equal to $(A + 4) \text{ units}^2$. The area between the curve $y = f(x)$, the x -axis, and the line $x = c$ is equal to $(2A - 4) \text{ units}^2$. Use this information to evaluate $\int_a^c f(x) dx$.



- (A) $A + B$ (B) $4A - 8$ (C) $2A + 8$ (D) $A - 8$

Section II		
1.	Find the primitive function for	
	(i) $\frac{3x^3+7x}{x^2}$	2
	(ii) $\frac{1}{\sqrt{4x-3}}$	2
2.	A is the point $(1, 5)$ and $B(6, -2)$. The point $P(x, y)$ moves so that PA is perpendicular to PB . Find the equation of the locus of P .	2
3.	If $p, q, p + q$ is in an arithmetic progression and $p, q, 20$ is in geometric progression, where $p > 0, q > 0$, find the values of p and q .	2
4.	The line $y = 2x - 8$ is a tangent to the parabola $x^2 = 4ay$ at the point P where $x = 8$	
		
	(i) Show that $a = 2$.	2
	(ii) Write down the coordinates of the focus and equation of directrix.	2
5.	Given the series $2^{x-3} + 2^{2x-4} + 2^{3x-5} \dots$	
	(i) Find an expression for T_{20} .	2
	(ii) For what values of x does the series have a limiting sum?	2
6.	The origin, O and the point P and Q are the vertices of the shaded area in the diagram. The sides lie on curves with equations $y = x(x + 3)$, $y = x - \frac{x^2}{4}$ and $y = \frac{4}{x^2}$.	
	(i) P and Q have the coordinates $(p, 4)$ and $(q, 1)$. Find the value of p and q .	2
	(ii) Calculate the shaded area.	3
		

7.	The diagram shows a cone of base radius r cm and height h cm inscribed in a sphere of radius 50 cm. The centre of the sphere is O and $\angle OAB = 90^\circ$. Let OA = x cm. (i) Show that $r = \sqrt{2500 - x^2}$. 1 (ii) Hence show that the volume, V cm³, of the cone is given by: 2 $V = \frac{\pi}{3} (2500 - x^2)(50 + x)$ (iii) Find the radius of the largest cone which can be inscribed in the sphere.(give your answer to nearest mm.) 3
8.	Mona is retiring next week and her Superannuation Fund contains \$1 200 000. The Fund is earning 6% p.a. compound interest, compounded monthly. Mona wishes to withdraw a regular amount of \$8 000 per month to live in her retirement. (i) Show that after 1 month she will have an amount A_1 in her account where $A_1 = 1\,200\,000(1.005) - 8\,000$ 1 (ii) Show that after 3 months the amount in her account A_3 is given by $A_3 = 1\,200\,000(1.005)^3 - 8\,000\{(1.005)^2 + (1.005) + 1\}$ 2 (iii) By finding a similar expression for the amount remaining after n months, find how many years the money will last. 3

~ END OF EXAM ~

Solutions yr 11 Dec. Task

Section I

M.C ① D ② C ③ C ④ D

Section II

Q1 (i) $\int \frac{3x^3 + 7x^2}{x^2} dx$
 $= \int (3x + 7) dx$ ①
 $= \frac{3x^2}{2} + 7x + c$ ①

ii) $\int \frac{1}{(4x-3)^{1/2}} dx$
 $= \int (4x-3)^{-1/2} dx$ ①
 $= \frac{(4x-3)^{1/2}}{2 \times \frac{1}{2}} + c$ ①
 ~~$\frac{1}{2} \times \frac{1}{x}$~~ ignore c
 $= \frac{1}{2} \times \sqrt{4x-3} + c$

Q2 $m_{AP} = \frac{y-5}{x-1}$
 $m_{BP} = \frac{y+2}{x-6}$ ①
 $\therefore m_{AP} \times m_{BP} = -1$ ①
 $\frac{y-5}{x-1} \times \frac{y+2}{x-6} = -1$

$y^2 + 2y - 5y - 10 = -(x^2 - 6x - x + 6)$
 $x^2 - 7x + y^2 - 3y - 4 = 0$

Q3 $p, q, p+q$ in A.P

$q - p = p + q - q$

$q = 2p$

$p, q, 20$ in G.P

$\frac{q}{p} = \frac{20}{q} \Rightarrow q^2 = 20p$ ①

$\therefore (2p)^2 = 20p$

$4p^2 - 20p = 0$

$4p(p-5) = 0 \Rightarrow p = 0, 5$ but $p > 0$

$\therefore p = 5$
 $q = 10$ ①

Q4 $m_T = 2$ — (A)

Also $x^2 = 4ay$

$y = \frac{x^2}{4a}$

$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$

$m_T = \left. \frac{dy}{dx} \right|_{x=8} = \frac{8}{2a} = \frac{4}{a}$ — B ①

From A & B

$2 = \frac{4}{a}$ — (1)

$\therefore a = 2$ Shown.

(ii) $F(0, 2)$, eq. of director $y = -2$ ①
 ①

Q5 (i) G.P with $r = \frac{2^{2x-4}}{2^{2x-3}}$

$= 2^{2x-4-x+3} = 2^{x-1}$ ①

$T_{20} = ar^{19}$

$= 2^{x-3} \times (2^{x-1})^{19}$

$= 2^{x-3} \times 2^{19x-19}$

$= 2^{20x-22}$ ①

1) $-1 < r < 1$
 Since $r = 2^{x-1}$
 $2^{x-1} > 0$
 $0 < 2^{x-1} < 1$
 $2^{x-1} < 2^0$
 $x-1 < 0$
 $x < 1$ — (1)

16 $y = \frac{4}{x^2}$
 $4 = \frac{4}{p^2} \Rightarrow p^2 = 1$ — (1)
 $p = 1 \rightarrow$ in the first quad.

$p(1, 4)$

$1 = \frac{4}{q^2} \Rightarrow q = 2$

$q = (2, 1)$ — (1)

$A = \int_0^1 x(x+3) - (x - \frac{x^2}{4}) dx + \int_1^2 (\frac{4}{x^2} - x + \frac{x^2}{4}) dx$
 $= \int_0^1 (\frac{5}{4}x^2 + 2x) dx + \int_1^2 (\frac{4}{x^2} - x + \frac{x^2}{4}) dx$

$= \left[\frac{5}{12}x^3 + \frac{2x^2}{2} \right]_0^1 + \left[-\frac{4}{x} - \frac{x^2}{2} + \frac{x^3}{12} \right]_1^2$

$= \left(\frac{5}{12} + 1 \right) + \left[\left(-\frac{4}{2} - \frac{4}{2} + \frac{8}{12} \right) - \left(-\frac{4}{1} - \frac{1}{2} + \frac{1}{12} \right) \right]$

$= \frac{17}{12} + \left[-4 + \frac{8}{12} - \left(-\frac{48-6+1}{12} \right) \right]$

$= \frac{17}{12} + \left[-\frac{40}{12} - \left(-\frac{53}{12} \right) \right]$

$= \frac{17}{12} + \left[\frac{13}{12} \right] = \frac{30}{12} = \frac{5}{2} = 2.5 \text{ sq. units}$

Q7 (i) In rt angled ΔOAB
 $50^2 = r^2 + x^2$
 $r = \pm \sqrt{2500 - x^2}$
 as $r > 0 \therefore r = \sqrt{2500 - x^2}$ — (1)

(ii) $V = \frac{1}{3} \pi r^2 h$
 $\therefore h = OA + AB = 50 + x$ — (1)
 $= \frac{1}{3} \pi (2500 - x^2)(50 + x)$

(iii) $\frac{dV}{dx} = \frac{1}{3} \pi [2500 - x^2] + (50 + x)x^{-2}$
 $= \frac{1}{3} \pi [2500 - x^2 - 2x^2 - 100x]$
 $= \frac{1}{3} \pi [2500 - 3x^2 - 100x]$

for max. & min. $\frac{dV}{dx} = 0$

$2500 - 3x^2 - 100x = 0$

$x = \frac{-100 \pm \sqrt{(100)^2 + 4 \times 3 \times 2500}}{6}$

$= \frac{-100 \pm 200}{6}$

$= -\frac{300}{6} \text{ or } \frac{100}{6}$

But x can't be negative — (1)
 $\therefore x = \frac{100}{6} = \frac{50}{3}$

$\frac{d^2V}{dx^2} = \frac{\pi}{3} (-100 - 6x)$

$= \frac{\pi}{3} (-100 - 100) < 0$

$\therefore$ max.

$\therefore r = 47.14$ — (1)

Q8 $6\% \text{ p.a} = 0.5\% \text{ p.m}$

$$(i) A_1 = 1200000(1.005) - 8000 \quad \text{--- (1)}$$

$$(ii) A_2 = A_1 \times 1.005 - 8000$$

$$= [1200000(1.005) - 8000] \times 1.005 - 8000$$

$$= 1200000(1.005)^2 - 8000(1 + 1.005)$$

$$A_3 = A_2 \times 1.005 - 8000$$

$$= [1200000(1.005)^2 - 8000(1 + 1.005)] \times 1.005 - 8000$$

$$= 1200000(1.005)^3 - 8000(1 + 1.005 + 1.005^2)$$

--- 2 marks with working.

$$(iii) A_n = 1200000(1.005)^n - 8000(1 + 1.005 + \dots + 1.005^{n-1})$$

$\downarrow$
 h.p
 $a=1, r=1.005$

Also $A_n = 0$.

$$\textcircled{1} \quad 0 = 1200000(1.005)^n - \frac{8000(1.005^n - 1)}{1.005 - 1} \quad \text{--- (1)}$$

$$1200000(1.005)^n = \frac{8000(1.005^n - 1)}{0.005}$$

$$5 \times 12 \times (1.005)^n = 80[(1.005)^n - 1]$$

$$(1.005)^n [60 - 80] = -80$$

$$\cancel{20} (1.005)^n = \cancel{80}$$

$$(1.005)^n = 4$$

$$\ln(1.005) = \ln 4$$

$$n = \frac{\ln 4}{\ln(1.005)} = 277.95 \text{ months}$$

$$= 23 \text{ years.}$$